

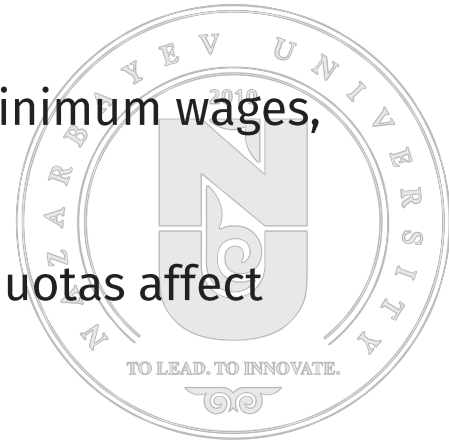
The Basics of Supply and Demand

**ECON 101: Introduction to
Microeconomics**

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The Basics of Supply and Demand

- Supply-demand analysis is a fundamental and powerful tool that can be applied to a wide variety of interesting and important problems. To name a few:
 - Understanding and predicting how changing world economic conditions affect market price and production
 - Evaluating the impact of government price controls, minimum wages, price supports, and production incentives
 - Determining how taxes, subsidies, tariffs, and import quotas affect consumers and producers



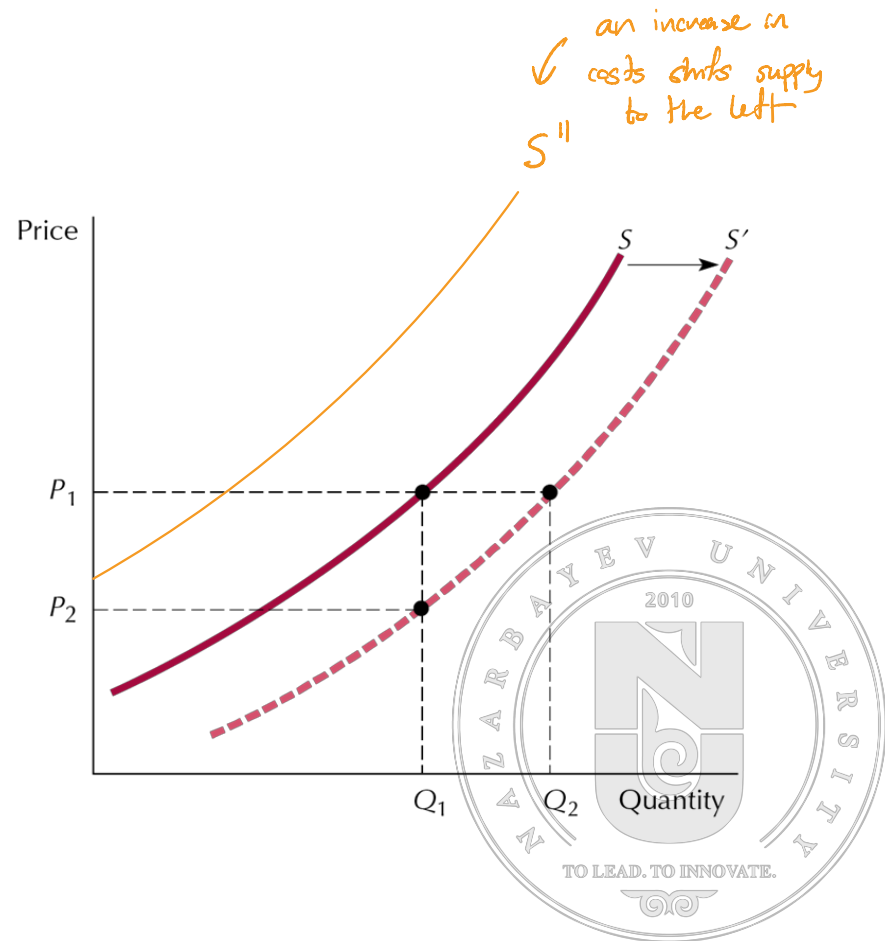
Supply and Demand

The supply curve

- The supply curve describes the relationship between the quantity of a good that producers are willing to sell and the price of the good.

$$Q_S = Q_S(P, DET^S)$$

- The supply curve, labeled S in the figure, shows how the quantity of a good offered for sale changes as the price of the good changes. The supply curve is upward sloping: The higher the price, the more firms are able and willing to produce and sell.
- If production costs fall, firms can produce the same quantity at a lower price or a larger quantity at the same price. The supply curve then shifts to the right (from S to S').



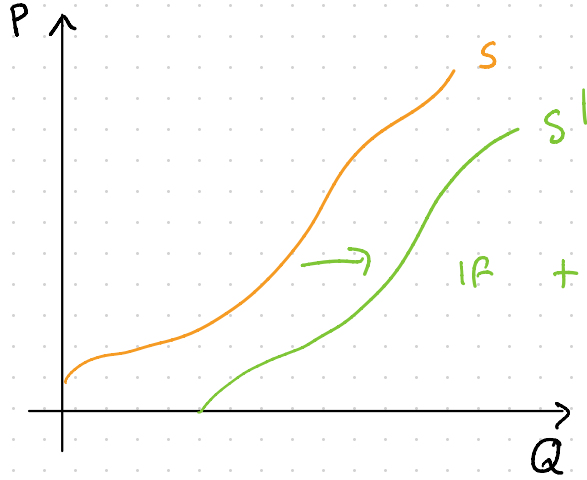
Supply and Demand

Other variables that affect supply

- The quantity that producers are willing to sell depends not only on the price they receive but also on their production costs, including wages, interest charges, and the costs of raw materials.
- When production costs decrease, output increases no matter what the market price happens to be. The entire supply curve thus shifts to the right.
- Economists often use the phrase change in supply to refer to shifts in the supply curve, while reserving the phrase change in the quantity supplied to apply to movements along the supply curve.



The supply curve can be of any shape:



Notice S is upward sloping: $\uparrow P \Rightarrow \uparrow Q^s$
Price positively related to quantity supplied
"Law of Supply"

IR + Det of supply $\uparrow \Rightarrow$ Supply curve shifts to the right.

Linear Supply Curves

$$Q_s = A_s + dP \quad \text{where } d > 0$$

A_s : determinants of supply / term encompassing all supply determinants

- Wages, rents
- Technology
- N_s : number of sellers
- P^F : future prices

Inverting the supply curve: $P = \frac{1}{d} \cdot Q_s - \frac{A_s}{d}$

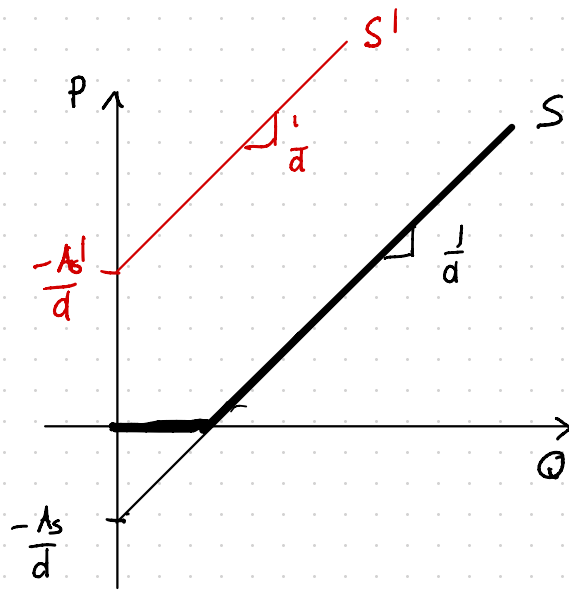
slope: $\frac{1}{d}$

intercept: $-\frac{A_s}{d}$

$$Q_s = 5 + 2P$$

$$\frac{Q_s - 5}{2} = P$$

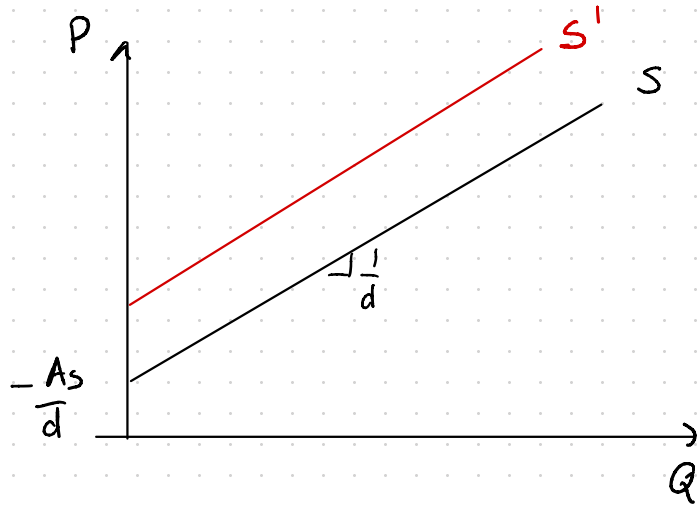
$$P = \max \left\{ 0, \frac{1}{d} \cdot Q_s - \frac{A_s}{d} \right\}$$



Example:

$$Q_s = dP - \underbrace{c\omega - aR}_{+ bT} \quad d > 0, c > 0 \\ a, b > 0$$

(determinants of the supply are written explicitly)
 $\Rightarrow A_s = bT - c\omega - aR$



suppose $\omega \uparrow \Rightarrow A_s \downarrow$

$\Rightarrow -\frac{A_s}{d} \uparrow$

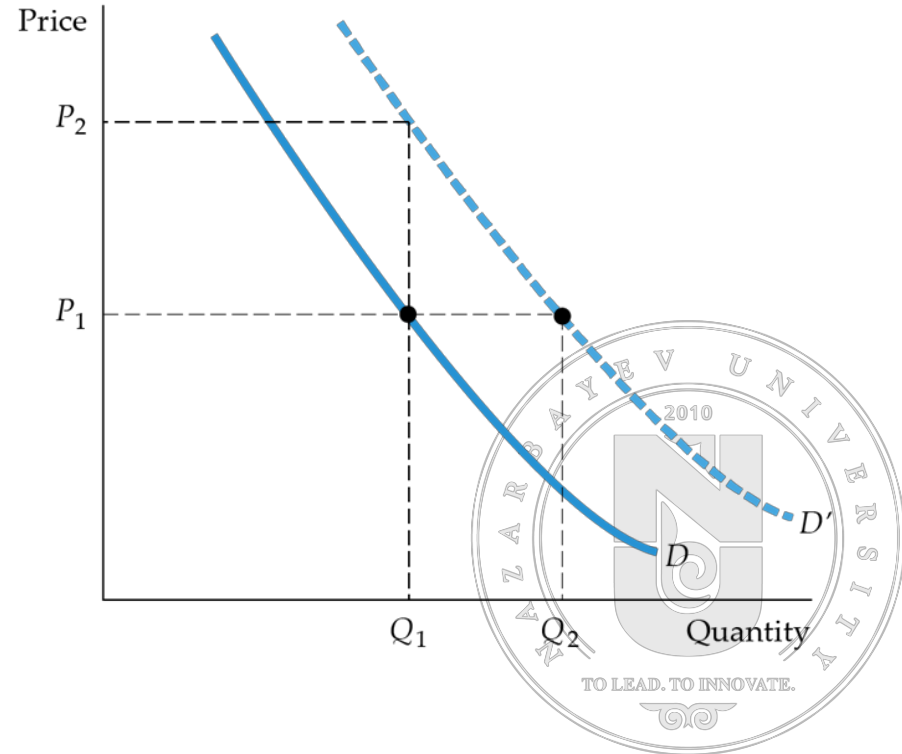
Supply and Demand

The Demand Curve

- The demand curve describes the relationship between the quantity of a good that consumers are willing to buy and the price of that good.

$$Q_D = Q_D(P, DET^D)$$

- The demand curve, labeled D , shows how the quantity of a good demanded by consumers depends on its price. The demand curve is downward sloping; holding other things equal, consumers will want to purchase more of a good as its price goes down.
- The quantity demanded may also depend on other variables, such as income, the weather, and the prices of other goods.
- For most products, the quantity demanded increases when income rises. A higher income level shifts the demand curve to the right (from D to D').



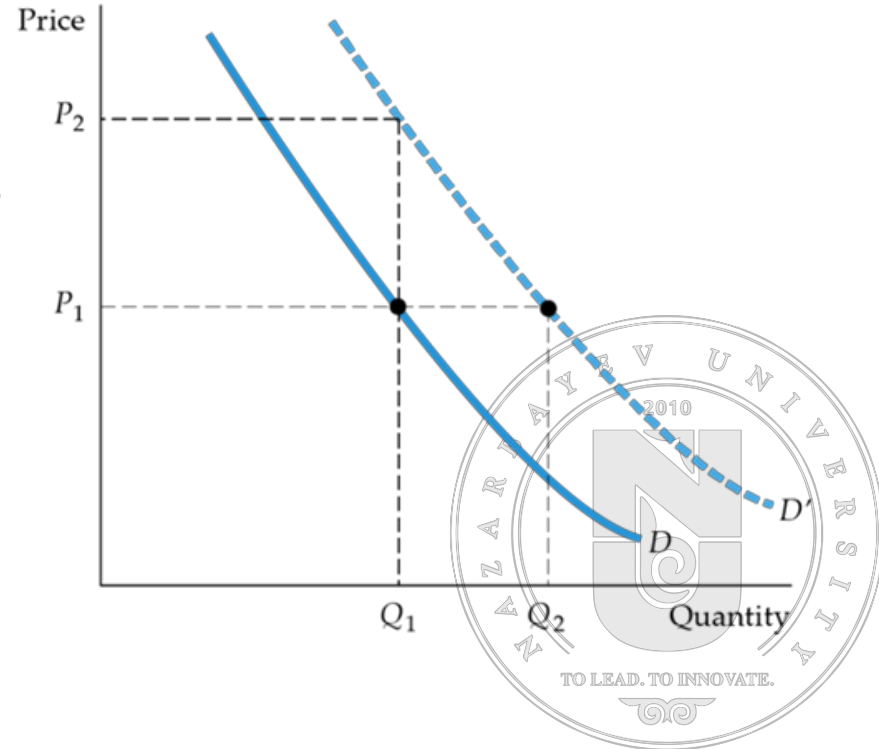
Supply and Demand

Shifting the demand curve

- If the market price were held constant at P_1 , we would expect to see an increase in the quantity demanded—say, from Q_1 to Q_2 , as a result of consumers' higher incomes. Because this increase would occur no matter what the market price, the result would be a shift to the right of the entire demand curve.

Substitutes and complementary goods

- **Substitutes** Two goods for which an increase in the price of one leads to an increase in the quantity demanded of the other.
- **Complements** Two goods for which an increase in the price of one leads to a decrease in the quantity demanded of the other.

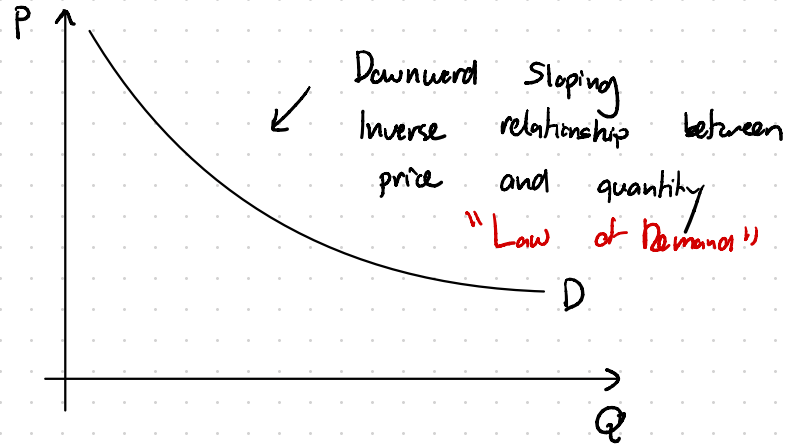


Determinants of Demand

$p^s \uparrow \rightarrow Q_0 \uparrow$
 $p^s \downarrow \rightarrow Q_0 \downarrow$ } + Shifters

$p^c \uparrow \rightarrow Q_0 \downarrow$
 $p^c \downarrow \rightarrow Q_0 \uparrow$ } - Shifters

Shape of Demand Curve



Linear Demand Curves

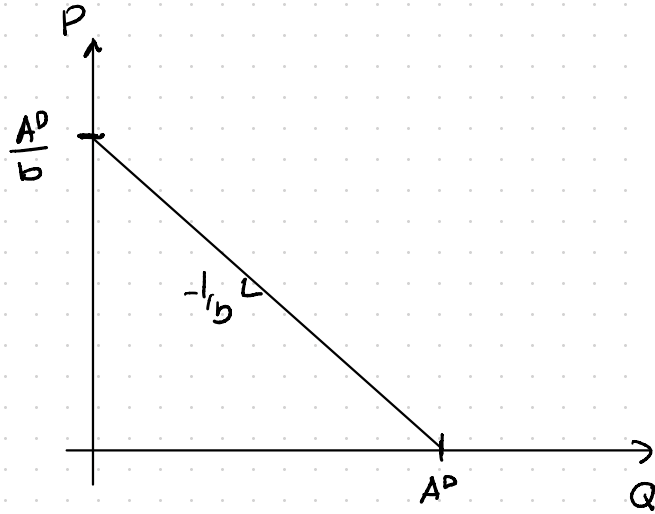
$$Q_D = A^D - bP, \quad \text{where } b > 0$$

To graph it, solve P :

$$P = \frac{A^D}{b} - \frac{Q_D}{b}$$

Intercept: $\frac{A^D}{b}$

Slope: $-\frac{1}{b}$



A^D : encapsulates the determinants of demand

- P^C , P^S

- Tastes & Preferences

- Future prices of the good.

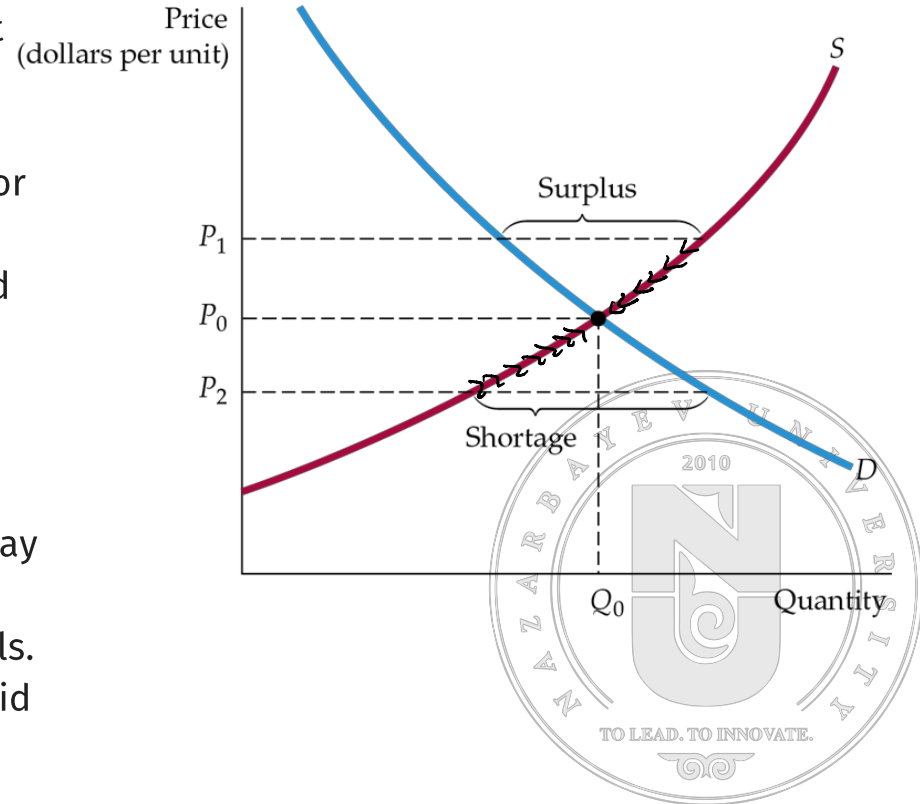
(Q_0, P_0) are the market equilibrium quantity and price.

The market mechanism

- **equilibrium (or market clearing) price** Price that equates the quantity supplied to the quantity demanded.
- **market mechanism** Tendency in a free market for price to change until the market clears.
- **Surplus** Situation in which the quantity supplied exceeds the quantity demanded.
- **Shortage** Situation in which the quantity demanded exceeds the quantity supplied.

The market clears at price P_0 and quantity Q_0 , so we say that the market is in equilibrium.

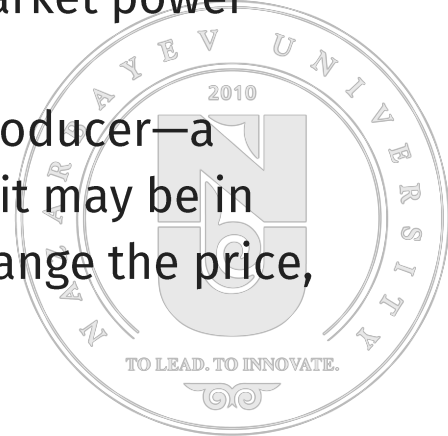
At the higher price P_1 , a surplus develops, so price falls. At the lower price P_2 , there is a shortage, so price is bid up.



The market mechanism

WHEN CAN WE USE THE SUPPLY-DEMAND MODEL?

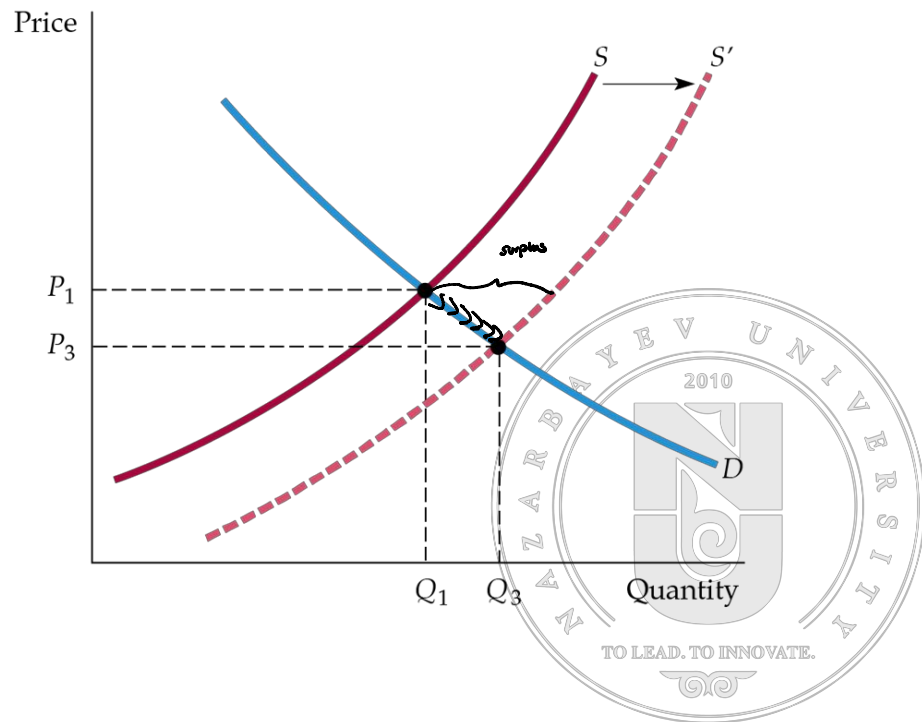
- We are assuming that at any given price, a given quantity will be produced and sold.
- This assumption makes sense only if a market is at least roughly competitive. By this we mean that both sellers and buyers should have little market power—i.e., little ability individually to affect the market price.
- Suppose instead that supply were controlled by a single producer—a monopolist. If the demand curve shifts in a particular way, it may be in the monopolist's interest to keep the quantity fixed but change the price, or to keep the price fixed and change the quantity.



Changes in market equilibrium

New Equilibrium following shift in supply

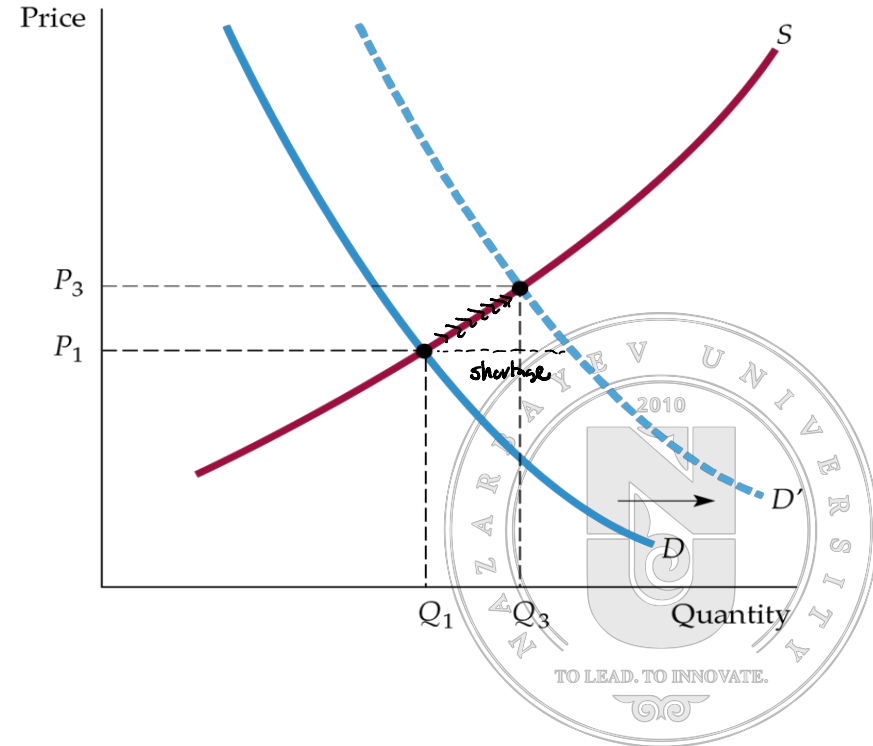
- When the supply curve shifts to the right, the market clears at a lower price P_3 and a larger quantity Q_3 .



Changes in market equilibrium

New Equilibrium following shift in demand

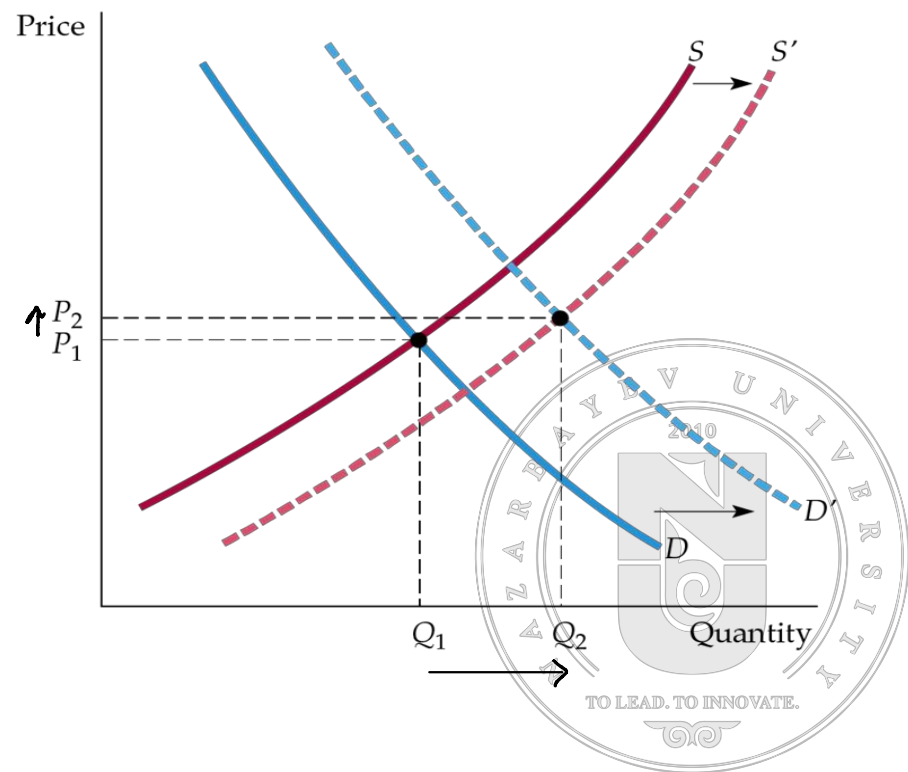
- When the demand curve shifts to the right, the market clears at a higher price P_3 and a larger quantity Q_3 .

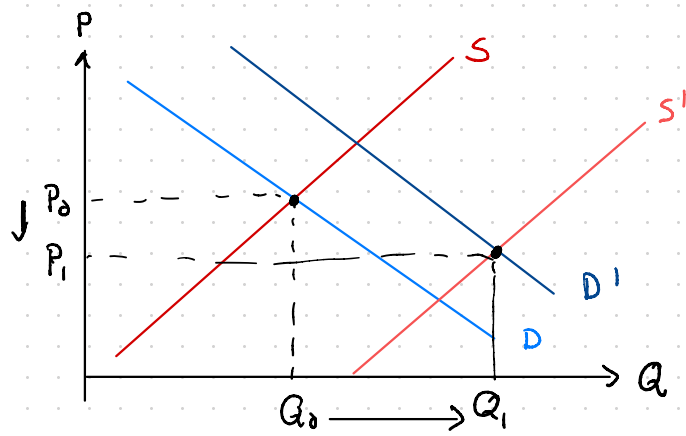


Changes in market equilibrium

New Equilibrium following shifts in supply and demand

- Supply and demand curves shift over time as market conditions change.
- In this example, rightward shifts of the supply and demand curves lead to a slightly higher price and a much larger quantity.
- In general, changes in price and quantity depend on the amount by which each curve shifts and the shape of each curve.



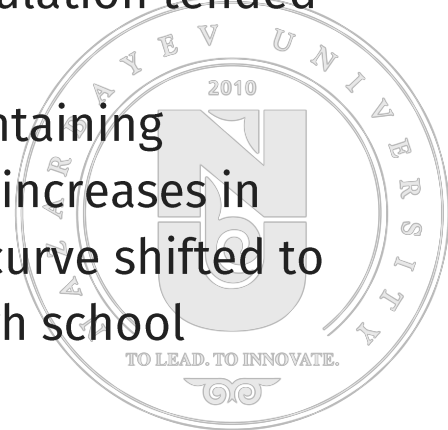


Moral: When both $D, S \uparrow$
 $\therefore Q^* \uparrow$, $P?$ depends on how much
each schedule (curve) moves relative to
the other.

Examples

The price of eggs and the price of college education

- From 1970 to 2010, the real (constant-dollar) price of eggs fell by 55 percent, while the real price of a college education rose by 82 percent.
- The mechanization of poultry farms sharply reduced the cost of producing eggs, shifting the supply curve downward. The demand curve for eggs shifted to the left as a more health-conscious population tended to avoid eggs. (not true anymore!)
- As for college, increases in the costs of equipping and maintaining modern classrooms, laboratories, and libraries, along with increases in faculty salaries, pushed the supply curve up. The demand curve shifted to the right as a larger percentage of a growing number of high school graduates decided that a college education was essential.

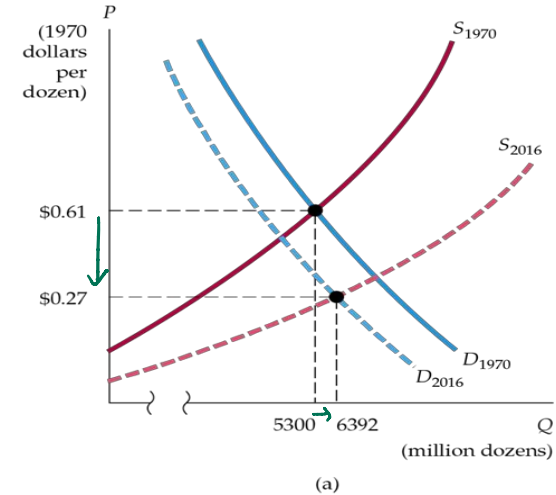


Examples

The price of eggs and the price of college education

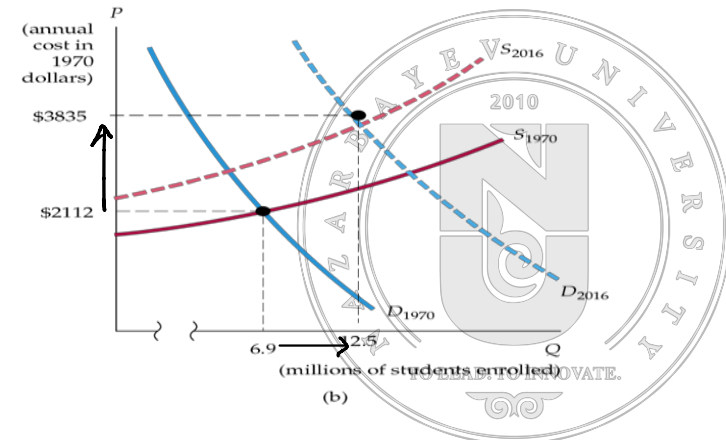
MARKET FOR EGGS (a)

- The supply curve for eggs shifted downward as production costs fell; the demand curve shifted to the left as consumer preferences changed.
- As a result, the real price of eggs fell sharply, and egg consumption rose.



MARKET FOR COLLEGE EDUCATION (b)

- The supply curve for a college education shifted up as the costs of equipment, maintenance, and staffing rose.
- The demand curve shifted to the right as a growing number of high school graduates desired a college education. As a result, both price and enrollments rose sharply.



Examples

Wage inequality in the US

MARKET FOR EGGS (a)

- Over the past two decades, the wages of skilled high-income workers have grown substantially, while the wages of unskilled low-income workers have fallen slightly.
- From 1978 to 2009, people in the top 20 percent of the income distribution experienced an increase in their average real (inflation-adjusted) pretax household income of 45 percent, while those in the bottom 20 percent saw their average real pretax income increase by only 4 percent.
- While the supply of unskilled workers—people with limited educations—has grown substantially, the demand for them has risen only slightly.
- On the other hand, while the supply of skilled workers—e.g., engineers, scientists, managers, and economists—has grown slowly, the demand has risen dramatically, pushing wages up.



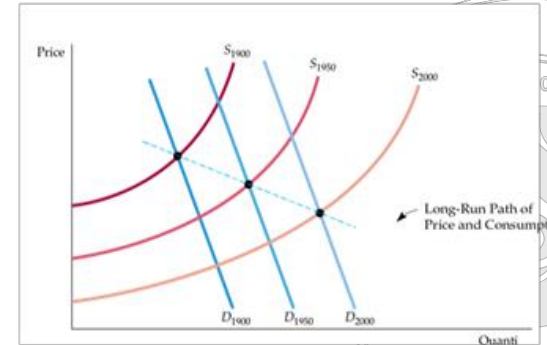
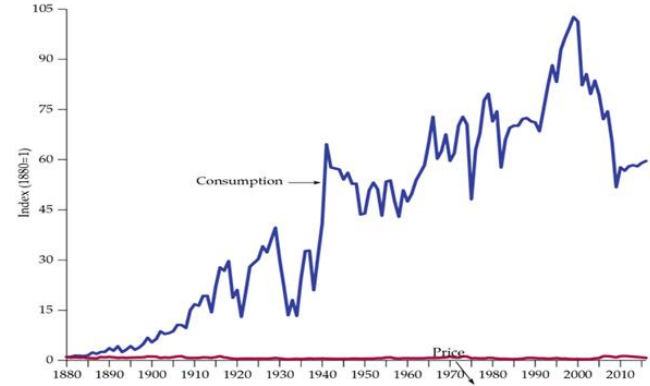
Examples

Long-run behavior of natural resource prices: consumption and price of copper

Although annual consumption of copper has increased about a hundredfold, the real (inflation-adjusted) price has not changed much.

Long-run, movements of supply and demand for mineral resources

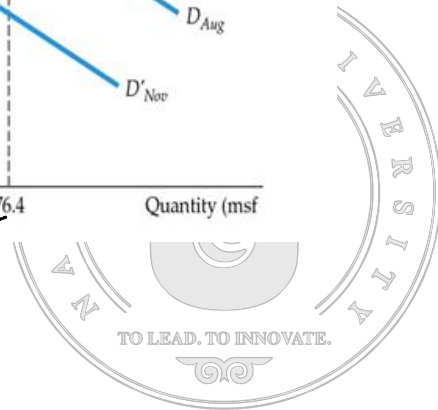
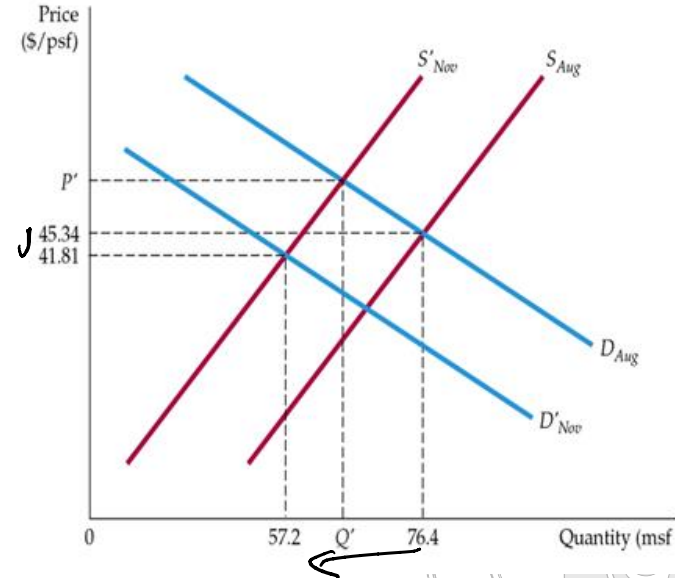
Although demand for most resources has increased dramatically over the past century, prices have fallen or risen only slightly in real (inflation-adjusted) terms because cost reductions have shifted the supply curve to the right just as dramatically.



Examples

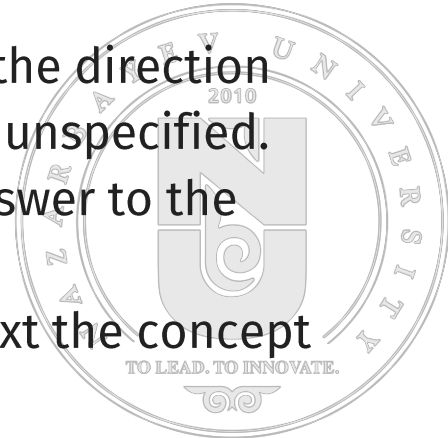
THE EFFECTS OF 9/11 ON THE SUPPLY AND DEMAND FOR NEW YORK CITY OFFICE SPACE

Following 9/11 the supply curve shifted to the left, but the demand curve also shifted to the left, so that the average rental price fell.



What's next?

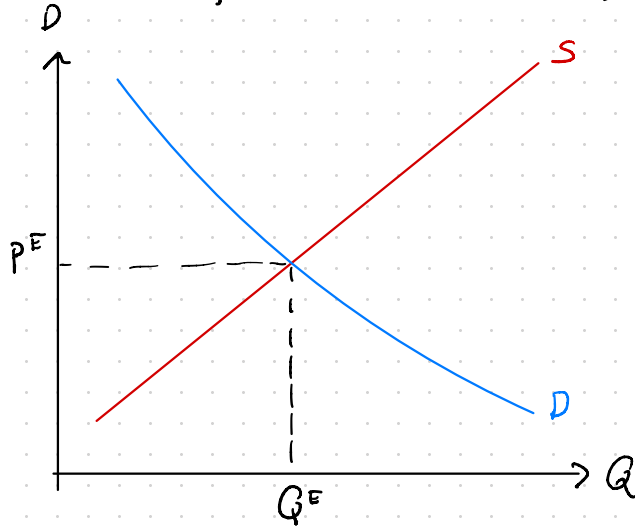
- Until now, we have only been able to give a qualitative answer when analyzing changes in market equilibria after a change in one of the determinants of either the supply or the demand.
- As we have seen, whenever just one of the determinants change, we can predict the direction of the change in the market equilibrium quantity and price.
- However, as soon as more than one determinant changes, the direction of the changes in one of the equilibrium variables remains unspecified.
- We need to be able to give a quantitative and definitive answer to the resultant market equilibrium.
- In order to give a quantitative answer, we will introduce next the concept of elasticity.



Market Equilibrium: Fix the determinants of D & S

Equilibrium: $Q_D(P^e) = Q_S(P^e)$

Market equilibrium (P^e, Q^e) for which the above eq. is satisfied



Surplus (excess supply): $P > P^e$

$$\text{Surplus} = Q_S(P) - Q_D(P)$$

Shortage (excess demand): $P < P^e$

$$\text{Shortage} = Q_D(P) - Q_S(P)$$

Linear Demand & Supply

$$Q_D = A^D - bP \quad b > 0$$

$$Q_S = A^S + dP \quad d > 0$$

For equilibrium we need that

$$A^D - bP^e = Q^e = A^S + dP^e$$

Solve for P^e

$$P^e = \frac{A^D - A^S}{d + b}$$

To obtain Q^e substitute into either Q_D or Q_S

$$Q^e = A^D - b \left(\frac{A^D - A^S}{d + b} \right)$$

$$= \frac{(d+b)A^D - b(A^D - A^S)}{d+b}$$

$$= \frac{dA^D + bA^S}{d+b}$$



End of this lecture.
Read Chapter 2 from Pindyck & Rubinfeld.